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Nut and Seed Products (Food Group 12)

Introduction

There are currently 133 food items in the Nut and Seed Products food group within the USDA National Nutrient Database for Standard Reference (SR). The sources of nutrient data for these items released in SR include the scientific literature, analytical studies, and food industry. In 1999 the International Tree Nut Council (INC) collaborated with the Nutrient Data Laboratory (NDL) to sample and analyze almonds, hazelnuts, macadamias, pecans, pistachios, and English walnuts. Since 2002, additional data were obtained via NDL's National Food and Nutrient Analysis Program (NFNAP) (p. 52) for those nuts as well as almond butter, Brazil nuts, cashews, pine nuts, shredded coconut, and mixed nuts; and flaxseed, pumpkin, sesame and sunflower seeds.

The Almond Board of California provided data on many varieties of almonds commonly grown in California. The Western Pistachio Association (now called American Pistachio Growers) provided data on both raw and dry roasted pistachio nuts. These industry data were then aggregated with the INC and NFNAP data. The remaining nut and seed nutrient data were derived primarily from the scientific literature.

Lightly salted mixed nuts (both with and without peanuts), lightly salted almonds, smoke-flavored almonds and glazed walnuts were added to SR due to their high sales volume. The data were derived from food labels and the food industry.

Apparent nutritional differences in estimates for different forms of nuts and seeds are not necessarily due to the effects of processing or preparation methods. Different forms of nuts and seeds (e.g., raw, oil roasted and dry roasted) usually were not of a common sample origin. Data were obtained from many sources and may represent different growing years, growing areas, cultivars, processing techniques, lengths and conditions of storage, laboratories, and possibly different methods of analysis. The above factors, together with natural variability, may lead to difference in nutrient content not related to the processing or preparation methods. The USDA ARS Germplasm Resources Information Network (GRIN) database (USDA, 2012) was used as the basic reference for scientific names and preferred common names.

Nuts

Several tree nuts are grown in the United States and are commercially important in the food supply. In fact, the United States produces over one-tenth of the world's tree nuts (USDA, 2012). Most coconuts consumed in the United States are imported, but in some tropical areas, coconuts are readily available from the coconut palm tree. Other nuts must be imported, such as Brazilnuts, cashew nuts, European chestnuts, ginkgo nuts, and pine nuts. Detailed information about tree nut imports and exports can be found through the USDA's Economic Research Service at www.ers.usda.gov. Relatively small amounts of some types of nuts are gathered by the consumer, either from the woodlands or from trees used to landscape property. Nuts in this category are

acorns, beechnuts, butternuts, Chinese and Japanese chestnuts, and hickorynuts. Black walnuts are gathered by families across 16 states and bought by nut companies to be shelled and packaged for sale to consumers (Hammons, 2007).

Nearly all nuts must be dried or cured prior to being consumed. Drying nuts to a low moisture content—either by sun drying or by mechanical drying—prevents deterioration of the nut and protects its quality during storage (California Walnut Board, 2011). Although coconuts can be eaten raw, directly from the tree, harvested green coconuts are partially dried or seasoned in the field (Grimwood, 1975). According to USDA's Economic Research Service (2010), the per capita consumption of tree nuts has increased from 2-1/4 pounds in the 1980's and 1990's to 3-1/2 pounds as of 2009.

Almonds (*Prunus dulcis*) make up over 1/3 of the per capita consumption of tree nuts in the US (USDA, 2010). About 80% of the world's supply and approximately 100% of the domestic supply are grown in the Central Valley of California. The market forms available include inshell almonds; shelled almonds in many unblanched and blanched forms (whole, sliced, slivered, chopped, and ground); almond paste made from ground unblanched or blanched almonds blended with sugar; almond butter made from ground dry roasted almonds blended with salt and a stabilizer; almond meal; and almond powder or flour (Almond Board of California, 2010).

Brazilnuts (*Bertholletia excelsa*) come from large trees that grow in the tropical jungles of the Amazon Basin. They are produced mainly in Bolivia, Brazil, and Peru. Both in-shell and shelled Brazilnuts are marketed (INC, 2008). These nuts are known for their very high selenium content, which is highly variable, depending on the geographic location where the nuts are found (Chang, 1995).

Cashew nuts (*Anacardium occidentale*), which are native to Brazil, are cultivated in many tropical countries, especially India, Vietnam, Mozambique and Brazil (INC, 2008). Before the cashew nuts are eaten, the corrosive liquid between the shells must be removed by some form of heat treatment (Woodroof, 1979), generally roasting. Raw cashews have been heat-treated to safely remove the kernel from the shell, but have not been further roasted.

Most **coconuts** (*Cocos nucifera*) eaten in the United States are consumed as dried (desiccated) coconut, a form of coconut meat that has been shredded, disintegrated, and then thoroughly dried in hot air driers (Grimwood, 1975). Unsweetened and sweetened coconut comes in many forms, called cuts. The two types of coconut cuts most often found in retail markets are flaked and shredded desiccated sweetened coconut (General Foods, 1982). Another type of desiccated coconut is toasted coconut which may be either unsweetened or sweetened (Ruehrmund, 1974).

Hazelnut and filbert are names used interchangeably for species of the genus *Corylus* (Hazelnut Marketing Board, 2014; Woodroof, 1979). Hazelnuts are grown commercially in Oregon and Washington, but filberts are also imported from Turkey, Italy, France, and Spain to meet the demand for this nut (Woodroof, 1979; INC, 2008). These nuts are

marketed in-shell and shelled, roasted or salted, and used for the preparation of several food products such as candy and ice cream (INC, 2008).

Macadamia nuts (*Macadamia integrifolia, M. tetraphylla*) are native to Australia. The *M. tetraphylla* variety has a rough shell which is not as good for roasting. The largest producers of macadamia nuts are Hawaii, Australia, South Africa, and Guatemala (INC, 2008). Dry roasted, salted macadamia nuts are readily available throughout the United States.

Botanically, **peanuts** and soybeans are legumes, not tree nuts. Thus, they are found in the Legume and Legume Products Food Group 16.

Pecans (*Carya illinoinensis*) are native to temperate North America, originating in central and eastern areas. The leading producers of pecans are Georgia and Texas, but they are also grown in several other states including Arizona, the Carolinas, Florida, and New Mexico, as well as Mexico. Over 80% of pecans sold have been shelled (NPSA, 2014).

One **pine nut** species, pinyon (*Pinus edulis*), is an important source of food and revenue for Native Americans in the Southwest. Very little recent information on the nutrient content of pinyons is available, except for the unpublished data of Lanner (1975) and Weber (1983). Most pine nuts (*Pinus* spp*.)* are imported from Italy, Spain, China, Portugal, and Turkey (INC, 2008). Pine nuts are marketed in shelled form and are generally used in the confectionery industry (INC, 2008) and as an ingredient in recipes.

Pistachio nuts (*Pistacia vera*) are cultivated in the United States, Iran, Turkey, Greece, Syria, and Italy. California is the major producer of pistachios in the US, with Arizona and New Mexico as additional sources. Pistachios are marketed primarily roasted and salted in their shell, but are also available unsalted and shelled (INC, 2008). Natural ivory-shelled and red-dyed pistachio nuts are available, but the percentage of dyed nuts is currently very small (American Pistachio Growers, 2011).

English walnuts (*Juglans regia*), which are often just called “walnuts,” originated in Persia (now known as Iran). Thus, they were first called Persian walnuts (Woodroof, 1979). After these walnuts were introduced to England and then brought to America they were called English walnuts (Brouk, 1975; Woodroof, 1979). Today, the Central Valley of California is the center of commercial production in the United States: 78% of the world’s supply and nearly all of the US production (California Walnut Board, 2011).

Black walnuts (*Juglans nigra*), are native to North America (Brouk, 1975). These walnuts, which are very hard to crack, are harvested wild from woodlands and from cultivated trees (Brouk, 1975, Woodroof, 1979) in the Midwest and East-Central part of the United States (Hammons, 2007).

Less Common Nuts

Acorns (*Quercus* spp.) are eaten raw, dried, or roasted in many parts of the world. American pioneers and native Americans ground acorns into meal to make bread or to thicken soups (Millikan, 1979). Most acorns contain potentially toxic tannins which must be leached out of the acorns before they are eaten. The literature contains very little recent information on the nutritive value of acorns. Weber, however, has made unpublished data available on the nutrient content of dried acorn kernels and full-fat acorn flour both of which are used by the White Mountain Apache Indians (Weber, 1983). Also see NDB No. 35182 for acorn stew which is an Apache dish.

Although **beechnut** trees (*Fagus spp*.) are found in many wooded areas, very few produce beechnuts that have a sweet flavor (Millikan, 1979). Beechnuts from these trees are difficult to gather for food because of their small size and poor nutmeat development.

Butternuts are another species of *Juglans* (*Juglans cinerea*). The trees are native to the Eastern United States and adjacent areas of Canada (USDA NRCS, 2011). The number of butternut trees in North America has decreased dramatically due to a canker fungal disease and other factors. The tree is considered an endangered species in Canada (Ministry of Natural Resources, 2009) and a species of special concern in all United States National Forests (Woeste and Pijut, 2009). Butternuts are used in baked products and candies due to their oily texture and nice flavor (Woeste and Pijut, 2009).

Almost all of the trees of the American **chestnut** (*Castanea dentata*) have been destroyed by fungus blight. Today, Chinese chestnut (*Castanea mollissima*) trees, which are blight resistant, are sold in place of the American chestnut for yard and orchard culture (Jaynes, 1979). Some Japanese chestnuts (*Castanea crenata*) are imported from Japan and some are grown in the United States, although these chestnuts are not as well adapted to the North American climate as the Chinese chestnuts (Jaynes, 1979). European or Italian chestnuts (*Castanea sativa*) are available in markets around the holiday season and are also sold roasted on street corners of some cities.

Dried **ginkgo nuts** (*Ginkgo biloba*) resemble almonds but are whiter, fuller, and rounder (Hedrick, 1972). Canned ginkgo nuts imported from Japan are readily available in Asian markets in the United States.

Hickorynuts, which are native to the woodlands of the United States, belong to the *Carya* species (Woodroof, 1979). Shagbark hickorynuts (*Carya ovata*) are moderate in size and thin shelled, while shellbark hickorynuts (*Carya laciniosa*) are larger nuts with a thick shell (Woodroof, 1979).

Pilinuts (*Canarium ovatum*) are imported from the Philippines. In the United States, markets featuring Philippine foods stock candied pilinuts.

Seeds

Seeds are grown primarily for their edible oils, because they have a very high fat content. Some seeds are eaten with very little home or commercial processing. Like nuts, some of the seeds are commercially important and can be easily purchased in retail or wholesale markets. Other seeds are available only to those having access to the growing plants or trees.

Both **pumpkin and squash seeds** (*Cucurbita* spp.) are consumed in the United States. Dried pumpkin and squash seeds and roasted pumpkin seeds are available in retail markets. Whole squash seeds are eaten roasted and salted by the Navajo Indians (Weber, 1983).

Safflower seed (*Carthamus tinctorius*) is grown in the United States – primarily California – as well as in Mexico, India, and the Middle East (US ITC, 2003). Although there has been some interest in using safflower seed meal and flour, safflower is cultivated primarily for oil in the seeds.

During the Civil War, **cottonseed** (*Gossypium* spp.) was parched and ground as a coffee substitute in the South (Hedrick, 1972). Today, glandless cottonseed products such as roasted kernels, flour, and meal are used as ingredients in a variety of products (Simmons, 1980) such as candy and baked products.

Sesame seed (*Sesamum indicum*) is native to East Africa and is grown in China, India, Ethiopia, Sudan, Nicaragua, Mexico, Guatemala, and the United States (Brouk, 1975). A paste form of sesame butter is made from the whole seed, while tahini, another type of sesame butter, is made from the kernel.

In the summer, **watermelon seeds** (*Citrullus lanatus*) are readily available from raw watermelons, but are probably seldom eaten from this source. Dried watermelon seeds, imported from Thailand and Taiwan, can be found in Asian markets in the United States.

Breadfruit trees (*Artocarpus altilis*) are found throughout the Tropics. Although most breadfruit trees bear fruit that is seedless, some cultivars contain seeds. In the seeded cultivars, the fruit pulp is almost nonexistent and the breadfruit seeds take up almost all the space inside the fruit (Brouk, 1975). Breadfruit seeds are boiled, roasted, or fried as a snack. They can also be ground into flour or used as nuts in baked products (FAO, 1989, South Pacific Commission, 1983).

In the American Tropics and Mexico, the **breadnut tree** (*Brosimum alicastrum*) produces yellow fruit with single seeds, also called ramóns (Peters and Pardo-Tejeda, 1982). The seeds are eaten raw or boiled and can be toasted and ground into a meal to make flatbread or a coffee-type beverage (Rocas, 2003).

One of the species of **Sisymbrium** (*Sisymbrium* spp.) is also known as tumble mustard. Native Americans of the Navajo nation use these dry ground seeds, which they call "k'ostse," as an ingredient of cornbread (Weber, 1983). Small-seeded plants that differ botanically from cereals, but that are cultivated like cereals in fields and ground into flour to make bread and similar products, are called pseudo-cereals (Brouk, 1975).

One of these pseudo-cereals is **chia seed** (*Salvia hispanica*) which is native to Mexico (Brouk, 1975).

The sacred **lotus** (*Nelumbo* spp*.)* is an aquatic plant found in China and India (Brouk, 1975). In the United States, Asian markets stock the dried, whole lotus seed which is imported from China. Traditionally, the seeds are roasted, candied, cooked in soup, and made into a paste for sauces and moon cakes (Dharmananda, 2002).

Cultivars of the confectionery type of **sunflower seeds** (*Helianthus annuus*) are generally black with white stripes and are larger than oilseed type cultivars (Adams, 1982). Whole sunflower seeds, sunflower kernels, and sunflower butter are sold.

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Pork Products (Food Group 10)

Introduction

Nutrient and food composition data for pork products are presented in the USDA National Nutrient Database for Standard Reference (SR). The data in SR represent the amount of each constituent in 100 grams of edible portion. The edible portion of pork may be represented as "separable lean and fat" or as "separable lean only". In each case, bone and connective tissue are removed from the cut and reported as refuse. In the case of "separable lean and fat", it is assumed that all fat present is consumed. For items described as "separable lean only", all external trim fat as well as trimmable seam fat are removed from the cut, and included in the reported refuse. Weights are determined for the whole retail cut as purchased, and for each component (e.g., refuse, separable lean, etc). The external trim fat and the seam fat are combined for analyses, weighed, and reported as separable fat. Nutrient analyses are conducted on the separable lean and the separable fat. The nutrient values for separable lean and separable fat are combined and weighted for their respective contributions to the whole retail cut; the resulting food items are reported as "separable lean and fat". For cooked pork cuts, the cuts are cooked with the separable fat intact. Nutrient data for separable fat, separable lean only, and separable lean and fat of cooked cuts are analyzed or calculated as described above.

The analytical nutrient data includes the mean nutrient value, the standard error given to three decimal places, and the number of observations on which the values are based. For many food items, mean values are given without an accompanying standard error and number of samples. These values are either calculated by pooling data or by weighting means, by applying cooking yields or nutrient retention factors to derive values for some cooked foods, or by imputation from a different, closely related food. For raw pork items and unheated cured items, nutrient values were calculated based on known content of the nutrient in the lipid (fatty acids), total solids (cholesterol), moisture-free, fat-free solids (minerals), or protein (water-soluble vitamins) fractions.

Nutrients

Nutrient information for SR can be found under "File Content" in the documentation. However, some nutrient information specific to pork products are included here. Nutrient values are obtained for moisture, protein, ash, and total fat. The values for protein are calculated from the content of total nitrogen (N) in the food using the conversion factor recommended by Jones (1941). The specific factor for protein applied to pork items is 6.25. The carbohydrate content of uncured products (except for some organ meats) consisting entirely of pork is negligible, and the carbohydrate content is thus assigned a zero value. The sum of the percentages of water, protein, total lipid, and ash do not necessarily equal 100 percent for those foods showing zero carbohydrate because the amounts of each of these constituents were determined independently.